Problem Set 1

Due Jan 16, 2026 at 11:59 PM Pacific

Turn in your problem set by Gradescope. Include any code you write in your solutions as well.

1. [10 Points.] Let's apply Bayes' Rule to a COVID test example. Let X be a random variable indicating whether or not a person has COVID, where $X = 1$ means they do have COVID, and $X = 0$ means they don't. The person has a low prior probability $p(X = 1) = 0.001$ of having COVID. Likewise $Y = 1$ indicates a positive COVID test result, and $Y = 0$ means a negative test result, but the test is not perfect. Suppose the test has a false negative rate $p(Y = 0 \mid X = 1) = 0.01$, and a false positive rate $p(Y = 1 \mid X = 0) = 0.1$.

   (a) [5 Points.] Suppose the person's first test is positive, $Y = 1$. Use Bayes' rule to find the posterior probability of the person having COVID, $p(X = 1 \mid Y = 1)$.

   (b) [5 Points.] Suppose the test is negative, $Y = 0$. Use Bayes' rule to find the posterior probability of having the illness, $p(X = 1 \mid Y = 0)$.

2. [10 Points.] Recall the definition of conditional independence: RVs Y_1 and Y_2 are conditionally independent given a third RV X iff $p(y_1, y_2 \mid x) = p(y_1 \mid x)p(y_2 \mid x)$.

   (a) [5 Points.] Prove that the above definition is equivalent to $p(y_1 \mid y_2, x) = p(y_1 \mid x)$ which is equivalent to $p(y_2 \mid y_1, x) = p(y_2 \mid x)$.

   (b) [5 Points.] Consider three independent random variables X, W_1, and W_2. Construct two new random variables $Y_1 = X + W_1$ and $Y_2 = X + W_2$. Are Y_1 and Y_2 independent? Are they conditionally independent given X? Prove your results.

3. [15 Points.] Consider again the person that may have COVID. Suppose the first test is positive for the illness, but the person wants a second opinion. Call the first test outcome $Y_1 = 1$, and the second test outcome Y_2. Assume X doesn't change between the two tests. Assume the measurements are conditionally independent, conditioned on the state X.

   (a) [5 Points.] Suppose the second test is also positive, $Y_2 = 1$. Use a recursive Bayesian estimator to find the posterior $p(X = 1 \mid Y_1 = 1, Y_2 = 1)$ given your result in Problem 1 above.

   (b) [10 Points.] How many consecutive positive tests will it take for the posterior probability of having COVID to be greater than not having COVID, i.e. what is n such that $p(X = 1 \mid Y_{1:n} = (1, \ldots, 1)) > 0.5$?

4. [15 Points.] As time passes, the person may become infected with COVID, or recover from it. Let $(X_1, \ldots, X_T)$ be the health state of the person on T consecutive days, and suppose X_t evolves as a Markov chain, with infection rate $p(X_{t+1} = 1 \mid X_t = 0) = 0.001$ and recovery rate $p(X_{t+1} = 0 \mid X_t = 1) = 0.05$. The student has one test per day. We will formulate a Bayesian filter to determine the probability of having the illness given a history of test results.

   (a) [5 Points.] To compress notation, let $\pi_t := [p(X_t = 0 \mid Y_{1:t} = y_{1:t}) \quad p(X_t = 1 \mid Y_{1:t} = y_{1:t})]^T$ be the belief vector at day t, let

   $$T := \begin{bmatrix} p(X_t = 0 \mid X_{t-1} = 0) & p(X_t = 0 \mid X_{t-1} = 1) \\ p(X_t = 1 \mid X_{t-1} = 0) & p(X_t = 1 \mid X_{t-1} = 1) \end{bmatrix}$$

   be the transition matrix, and

   $$M(y_t) := \begin{bmatrix} p(Y_t = y_t \mid X_t = 0) & 0 \\ 0 & p(Y_t = y_t \mid X_t = 1) \end{bmatrix}$$

   be the measurement likelihood matrix. Write down the recursive Bayesian filter in matrix form using only $M(y_t)$, T, π_{t-1} and π_t. (Hint: Feel free to use a vector of all ones, $\underline{1}$, if it will simplify your notation.)

(b) [10 Points.] On the 99th day, if the person has had 98 negative tests, with a positive test on day 99, what is the probability they have the illness? Assume the person started healthy with certainty $\pi_0 = [1 \quad 0]^T$. (Hint: write a short script to compute this.)